Shuting Pang

Medical Foundation Models · Multimodal Learning · Interpretable AI
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EDUCATION

University of Electronic Science and Technology of China (UESTC) Project-985 Chengdu, China
Master of Science, Computer Science Sep. 2024 to present

- GPA: **3.95**/4.0
- Awards: 2025 Sichuan Province Quality A-level Certification, 2025 Oppo Scholarship (Top 3%), 2025-2026 First-class Scholarship of UESTC.

Sichuan University (SCU) Project-985 Chengdu, China
Bachelor of Science, Computer Science Sep. 2020 to Jun. 2024

- GPA: **3.84**/4.0
- Awards: 2021 **National Scholarship** (Top 1%), 2022 Tencent Scholarship, 2021-2023 First-class Scholarship of SCU, 2024 Outstanding Graduate of Sichuan Province.

MAIN RESEARCH EXPERIENCES

Cross-Age Visual Foundation Models Transfer for Adolescent Mental Health Sep. 2025 to Present
Supervised by Prof. Xiaoshuang Shi(UESTC), in collaboration with Prof. Wei Zhang(West China Hospital)

- Developed a domain-aware semi-supervised framework combining adult clinical supervision with unlabeled adolescent community videos to address scarce adolescent clinical labels.
- Discovered depression-associated visual patterns across interview questions, temporal segments, and anatomically defined facial regions.
- Improved adolescent depression AUC by 6.61% over adolescent-only training, with further transfer to suicidal behavior prediction and few-shot community screening.

FITEL: Interpretable LLM Adaptation for EHR-Based Prediction Sep. 2024 to Jun. 2026
Supervised by Prof. Xiaoshuang Shi, in collaboration with Prof. Huan Song(SCU) | publication: [1]

- Built an automated EHR-to-tabular pipeline and developed learnable feature embeddings to adapt pretrained language models to structured clinical data without text serialization.
- Designed a dynamic sparse graph to jointly learn feature dependencies and select informative variables, reducing redundancy while identifying clinically meaningful predictors.
- Achieved the highest AUROC and AUPRC across three clinical tasks, improving AUROC over TabPFN by 0.57–1.77%, including 77.63% AUROC on the pain management task.

HA-CSL: Dual-Branch Consistency Learning for Interpretable MRI Classification May. 2023 to May. 2025
Co-supervised by Prof. Xiaoshuang Shi and Prof. Kang Li(SCU) | publication: [2,3]

- Developed a dual-branch MIL framework integrating 2D patch features and 3D volumetric context to reduce spatial information loss.
- Designed hierarchical attention and cross-branch consistency learning to identify salient features and align attention semantics.
- Improved classification accuracy by 4.4%, outperforming state-of-the-art methods in classification and interpretability on two MRI datasets.
- Contributed to the attention consistency objective in LCA-DB to reduce semantic discrepancies between image- and patch-based branches, extending cross-branch attention alignment to two more MRI datasets.

PUBLICATIONS

[1] S. Pang, Y. Dong, et al. "Feature-interpretable disease prediction from tabular data via dynamic GCN and LLMs," Pattern Recognition (2026). *[Link], IF 7.9.*

[2] S. Pang, Y. Chen, et al. "Interpretable 2.5D network by hierarchical attention and consistency learning for 3D. MRI classification," Pattern Recognition (2025). *[Link], IF 7.9.*

[3] R. Wang, X. Shi, **S. Pang**, et al. "Cross-attention guided loss-based deep dual-branch fusion network for liver. tumor classification," Information Fusion (2025). *[Link], IF 17.9.*